

ECE 50877
Introduction to Applied Cryptography
Project Proposals

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Proposal 1

Title: TPM for Linux Authentication

Problem Definition: Linux does not have a provider for TPMs.
We want to provide a way to unlock a master password via PIN.

Proposed Methodology:

- We will evaluate the use of hardware security modules on other mainstream operating systems, and primarily on Windows, the TPM, as it most cleanly maps to a Linux solution.
- We will identify current shortcomings and enumerate some attacks against the out-of-the-box Linux experience
- We will then investigate the authentication facilities on Linux, and determine if a TPM-based solution can fit within the constraints of those facilities, or whether a new userspace API is required.
- We will then investigate the interface provided by modern TPMs, and define an interface between the TPM and a userspace API. Then we will implement both pieces. We deliberately do not attempt to solve every bug in this implementation, and instead plan to produce a proof of concept.
- We will then test the previously enumerated attacks against the new implementation.

Expected Findings: We hope to document how other mainstream operating systems manage user authentication and secrets. We believe that by securing keys in a hardware security module, we will be able to defend against certain attacks without drastically impacting the user experience, and unlock a new avenue for user applications to store secrets and run-time.

Proposal 2

This proposal originates with Dr. Zahra Ghodsi of Purdue University.

Title: Differential Timing Analysis for Cryptographic Library Fingerprinting

Problem Definition: The starting step of exploitation is determining the target’s technology stack. An important part of this process is identifying the cryptographic libraries and implementations running on the target. Existing tools can often identify software through explicit version information or protocol behavior, but such information may be unavailable or intentionally hidden. This motivates the following research question: Can differential timing analysis identify the cryptographic library and version used by a remote target, despite network and system noise?

Proposed Methodology:

- We will construct a testbed containing multiple implementations and versions of commonly used cryptographic libraries.
- We will first collect timing measurements while each implementation performs a controlled suite of cryptographic operations under different inputs and configurations.
- We will analyze the resulting timing distributions to identify features that distinguish implementations.
- We will then train and evaluate statistical and machine-learning classifiers using these timing features.

Expected Findings: We expect the number of samples required to train accurate classifiers to rise substantially with noise. We also expect machine learning classifiers to outperform statistical classifiers, indicating broader applicability of machine learning techniques to side-channel analysis.